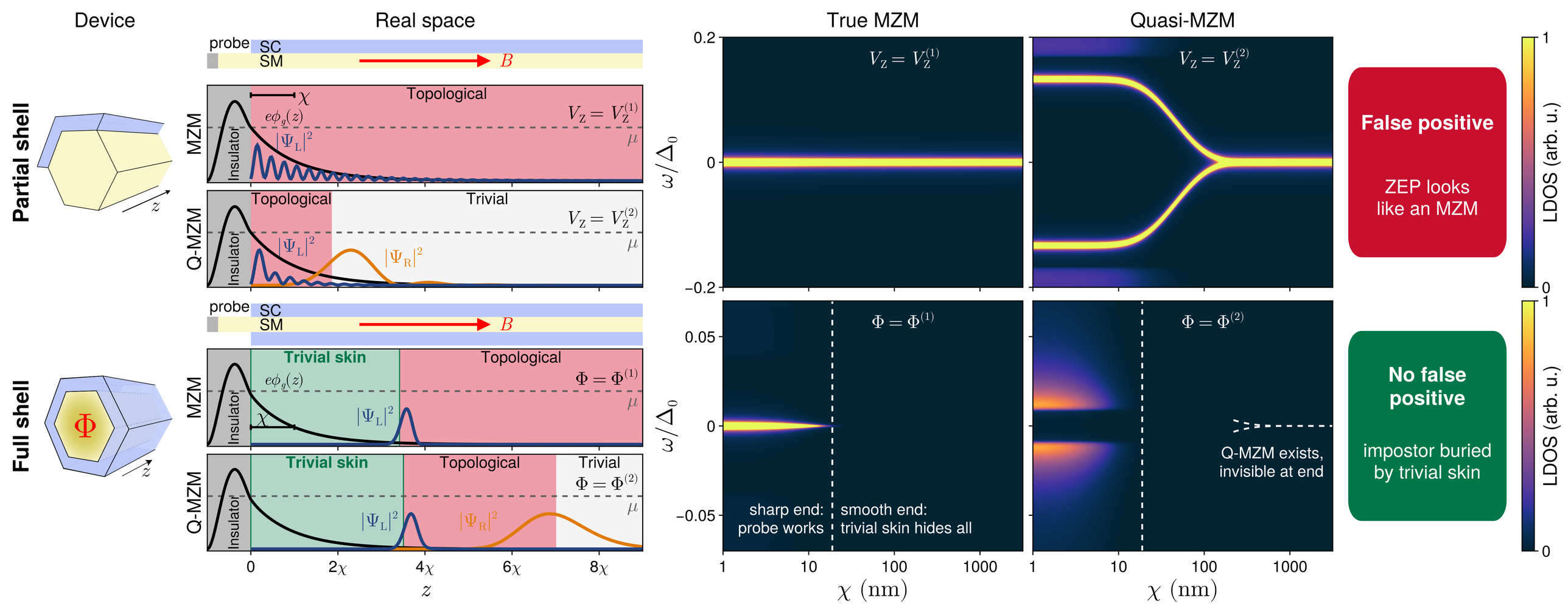

Device
Real space
True MZM
Quasi-MZM
Partial shell
Full shell
probe
SC
SM
B
MZM
Q-MZM
Insulator
Topological
Trivial
Trivial skin
χ
$e\phi_g(z)$
$|\Psi_L|^2$
$|\Psi_R|^2$
μ
$V_Z = V_Z^{(1)}$
$V_Z = V_Z^{(2)}$
$\Phi = \Phi^{(1)}$
$\Phi = \Phi^{(2)}$
Φ
z
0
2χ
4χ
6χ
8χ
ω/Δ_0
0.2
0
−0.2
0.05
−0.05
1
10
100
1000
χ (nm)
sharp end:
probe works
smooth end:
trivial skin hides all
Q-MZM exists,
invisible at end
False positive
ZEP looks
like an MZM
No false
positive
impostor buried
by trivial skin
LDOS (arb. u.)
1
0